

Blockchain - Based Land Registration For Agriculture Land

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Abstract - The proposed system introduces a secure and blockchain-based land registration structure designed specifically for agricultural land. Traditional land register systems depend on centralized database, causing them the risk of fraud, computer programs and unauthorized changes. To solve these challenges, we integrate blockchain and AI technologies to ensure a transparent, irreversible and fraudulent land registration process. Systems AI models such as Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Extreme Gradient Boosting (XGB), and easy-to-use gradient boosting machine (LGBM) to classify land items as cheating or non-cheating. Only confirmed non-Dhokhadhadi-land is stored in data blockchain networks, reduces computational overhead and increases efficiency. Smart contracts are distributed to validate the Land Registers in Spolia Test Network, where Solidity Analyzer ensures security and vulnerability evaluation. Non-Dhokhadhadi Landdata is safely stored in the Inter Planetary File System (IPFS), while the hash data is recorded on the Irreversible ledger of the blockchain for integrity and safety. The performance of the system is evaluated by the use of AI-accuracy matrix (ROC, Log-Loss score), blockchain scalability, gas cost use and bandwidth efficiency. By detecting AI-controlled scams with decentralized safety for blockchain, eliminates The proposed system addresses national fraud, enhances transparency, and builds trust in agricultural land registration. By detecting AI-driven fraud through the decentralized security of blockchain, it increases transparency and fosters confidence among stakeholders, including farmers, government officials, and financial institutions, ultimately streamlining the verification process.

INTRODUCTION

Advanced technologies including precision farming, crop monitoring, and intelligent manufacturing have significantly transformed land use, thereby driving sustainable land management by maximizing resource utilization, enhancing profitability, and reducing environmental damage. At the core of this change is a secure and efficient land registry system, since traditional central methods, typically kept by government agencies, are ineffective, suspect, and risk-sensitive. By cutting off intermediaries, boosting trust in property transactions, and enabling legal transfer of land rights without difficulty, blockchain technology offers a decentralized, transparent, and tamper-proof answer for these issues. Our recommended Blockchain-Enabled Land Registration System marries artificial intelligence and blockchain to give a very secure and fraud-resistant land ownership solution for farming. To identify counterfeit files before recording it on the blockchain and therefore guarantee storage of only verified and authentic records, uses of artificial intelligence models including logistic regression, support vector machines, and random forests are used. This supports data accuracy by reducing the superfluous computational load. Furthermore automating verification of land data, smart contracts drive accuracy, efficiency, and security in the registration process. Land records are kept securely on the Interplanetary File System (IPFS) and only the data's hash is put to the blockchain to provide more protection and flexibility to the system. This provides for openness without risk of unsanctioned changes. Before they are deployed on the Ethereum blockchain, vulnerability scanning of smart contracts using technologies like Slither help to guarantee a secure and steady execution. Reducing computational overhead and

blockchain congestion helps the proposed system also scales better by making transactions faster and less expensive. AI also can help discover discrepancies in real estate deals, therefore alerting potential misconduct before seriously damaging the system. The collaboration of AI and blockchain produces a clear and tamper-proof framework that helps to build trust among the many stakeholders, including legislators, farmers, and financial institutions. Moreover, the system benefits landowners by simplifying the proof process, reducing government red tape delays, and speeding up property transfer. Eliminating the middle requirement significantly reduces transaction costs and makes property registration easier and more affordable for little farmers. Expected to reduce land conflict, prevent illegal claims, and increase legal ownership, therefore advancing rural development and agricultural sustainability is the outcome of this approach. The Blockchain-Enabled Land Registration System improves transparency, efficiency, and overall effectiveness security in managing agricultural lands. By means of unchangeable record-keeping via blockchain and fraud detection via AI, it offers a future-proof answer to long-standing land registration issues. In template form for Microsoft Word, these revolutionary solution assures that land owners, agriculture stakeholders, and legislators have access to a secure, fraud-proof, and efficient land management system that enables long-term sustainability and economic growth in the farming industry. These are the instructions in template format for Microsoft Word and provide you with the general guidelines for preparing papers for ProComm 2024, July 14 - 17.

LITERARY RESEARCH

[1] Batras et al.(2022) recommend a decentralized land registry model to avoid fraud in property deals and enhance safety. The system offers immutability and openness by means of blockchain technology. The studies, however, lack artificial intelligence-based fraud detection and scaling issues remain.

[2]. Sarma et al. (2023) show a safe land registry system created with ethereum-based smart contracts. Sarma et al.'s efforts. Centers on blockchain transparency and reliability for land title registration. However, the poor capacity to identify fraudulent transactions results from the fact that the work excludes fraud detection tools.

[3] Tanwar et al. (2023) Use artificial intelligence models including SVM, RF, and XGB to spot questionable land transactions. Their research using machine learning methods help stop frauds. Still, the system is limited in preserving immutability of transaction histories by the lack of blockchain integration.

[4] agarwal et al. (2022) utilize blockchain and decentralized storage via ipfs for designing a system of land record management. This guarantees that data is handled securely and efficiently. Without artificial

intelligence-based fraud detection, though, the research would strengthen the system.

[5] ali et al. (2023) suggest a zero-knowledge proof dependent land registration system meant to protect privacy. Their work shows how blockchain can maintain anonymity yet be transparent in transactions. Still, the computational price of zero-knowledge proofs gives difficulties in actual deployment.

[6] Naguj et al. (2024) introduce the system "green land" where blockchain meets artificial fraud detection for secure land recordation. Smart contracts and ipfs provide scalable off-chain storage in this study. The system has computational overhead causing by artificial intelligence processing, strong or otherwise.

[7] Dubey et al. By means of Ethereum blockchain and smart contracts, they have suggested a secure land registration management system. It automates transactions and improves openness in the property dealings. Nevertheless, the emphasis of this paper appears to lie rather on execution than on giving a thorough examination of available and current solutions.

[8] Khalidi et al. Put forward a Split the data into an 80% training set and a 20% testing set. Use the training setA conceptual framework for blockchain-based land registration seeks to improve security and transparency. The research presents a theoretical basis for using blockchain technology in land registration but does not provide details on implementation in the real world., so its practical viability is hard to determine.

[9] Stefanovic et al. (2022) offer a land administration system based smart contracts to better security and ease dealings. The research is centred around automation and efficiency in property record management. The work brings to light slability and legal questions that might slow mass adoption disclaimed.system model and problem formulation. Here, we present a novel land registry system model by combining blockchain and AI to improve data security and integrity. Conventional land registry systems rely on centralized databases, denoted as C , which are managed either by one authority (a) or a set of authorities $\{a_1, a_2, \dots, a_k\} \in A$, in charge of maintaining land ownership records.

SYSTEM MODEL AND PROBLEM FORMULATION

1. Traditional Land Registry Model

A centralised land records system can be mathematically expressed prevalent:

C belongs to D and is a subset of the set last presented. Certainly not A but rather C .

where:

- D is the data collection containing land records.
- N users are in the system; every user is linked to a specific collection of land records:

$$\{U_1, \dots, U_i, \dots, U_m\} \rightarrow \{D_1, \dots, D_i, \dots, D_n\}$$

For all U_i , $1 \leq i \leq N$ let D_i .

U contains n ; D contains d .

2. Threat to security in center-based systems

Being centralized, the traditional land registry system is open to security breaches. A rogue agent (α) can change land records by exploiting security flaws.

$$A \in \{A \text{ OR } A_n\}$$

$$A \Rightarrow C.$$

$$\alpha \rightarrow \{d_{\{1\alpha\}}, d_{\{4\alpha\}}, d_{\{8\alpha\}}, d_{\{Fa\}}\} \text{ belongs to } D\alpha.$$

3. Suggested solution using artificial intelligence and blockchain-based systems

We propose an artificial intelligence-driven fraud detection module connected with blockchain technology to address these risks. An artificial intelligence classifier verifies the accuracy of a transaction prior to it being stored on the blockchain. Using supervised learning techniques, the classifier rates transactions as non-fraudulent or fraudulent.

$$O = \text{MAX } S_i, N_{\{1 \leq i \leq N\}} \times A$$

4. Training of an artificial intelligence model for fraud identification

Optimal performance for a AI classification model depends on a well-arranged data set (d):

$$D = (X_i, Y_i)_{\{i=1\}^N}$$

WHERE $Y_i \in \{0,1\}$, WHERE 0 = LEGITIMATE TRANSACTION, 1 = FRAUDULED DEALINGS

$X_1, \dots, X_i, \dots, X_n$ corresponds to $T_1, T_i, \dots, T_n$

Given these either I will $\rightarrow X$, $1 \leq i \leq N$.

5. Model optimization and fraud discovery

The data is divided into training and validation sets to ensure the AI classifier is trained in the most effective way. The training set helps the AI model learn transaction patterns, while the validation set helps estimate model performance without risk of overfitting.

$$L(\hat{y}, y) = - (1/n) \sum [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] .$$

6. Blockchain integration for advanced privacy levels.

The blockchain records only valid transactions once the AI model verifies one. This is how fraudulent transactions so never end up on the blockchain, hence improving data security and integrity.

PROPOSED METHODOLOGY

By including sophisticated functionality for further robust land enrollment, better performance, and more general use throughout husbandry and other sectors, the suggested system improves the current system result. Fraud discovery will be enhanced and more complicated data will be handled by AI systems like XGB and LGBM. A real-time monitoring result will identify and notify officers of possible fraud, thereby guarantee briskly response. Going forward, dynamic smart agreements will acclimate with new regulations and better data confirmation. delicacy will ameliorate with interoperability with geospatial

information systems and outside land registries. Rising quantities of data will be taken care of by better scalability features including sharding. Decentralized identity administration provides coproprietors with sequestration and security. More sophisticated vulnerability surveying systems will ameliorate security of smart contracts. Also supporting sustainable land operation is artificial intelligence. For bettered stoner experience and simple access to land records, the system will have an intuitive interface. These developments guarantee a land enrollment system that's further secure, effective, and scalable. Along with real-time monitoring, the suggested system provides stronger security and fraud discovery by means of sophisticated AI models including LGBM and XGB, thus enabling further nippy intervention. By means of dynamic smart contracts and comity with outside databases, it offers further inflexibility and further delicacy in land records. By their scalability and decentralized identity operation, the system guarantees strong performance, sequestration, and a smooth client experience for both officers and coproprietors.

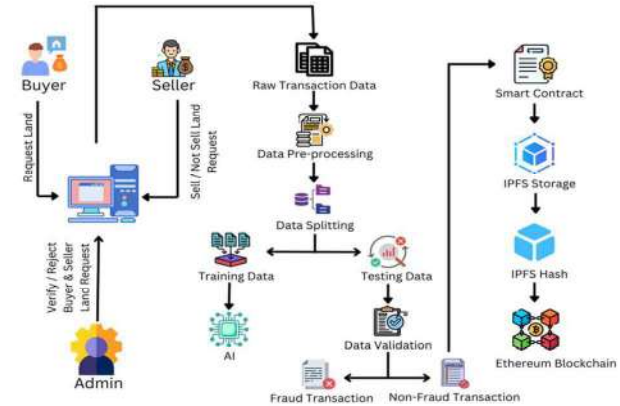


FIGURE 1. SYSTEM ARCHITECTURE

METHODOLOGY

Identification and analysis of issues

The approach first uses a thorough identification of the key issues traditional land registration systems face. These issues—theft, lack of openness, and inefficiency—that undermine the honesty and accuracy of land ownership records. A methodical examination of existing systems is conducted so that one can appreciate their deficiencies in an attempt to address these problems. This evaluation stresses the necessity of a more secure and efficient approach using artificial intelligence (AI) and blockchain systems to enhance the general efficiency of land registration processes.

System architecture design

The recommended approach is built into three main levels to support its operation. The initial layer, called the Data Layer gathers raw land ownership transaction data, including essential details on buyers, sellers, and parcels. The AI Layer, the second level, applies various machine learning techniques to classify land transaction data as either fake or not fake. Proper classifier is achieved with algorithms such as Extreme Gradient Boosting (XGB), Random Forest (RF), Support Vector Machine (SVM), Logistic Regression (LR), and Light Gradient Boosting Machine (LGBM). Finally, the Blockchain Layer stores confirmed transactions in a blockchain that is securely maintained, creating smart contracts to automate verifications and registration of land ownership information, therefore enabling for openness and immutability.

Data Collection and Preprocessing

In the approach, data preprocessing and acquisition are the main activities. One gets a usual set of land registry data including older transaction records needed to train the artificial intelligence models. The data set is cleansed to remove duplicates, errors, and extraneous information; missing values are dealt with using appropriate imputation techniques. Significant traits are then selected using ANOVA and Chi-squared tests, which help to improve the accuracy of the AI models. Data normalization is also done to offer feature uniformity, a major need for effective model training and improved classification accuracy.

AI Model Design

Over the AI model's creation, the clean data is Divide the data into an 80% training set and a 20% testing set. The training set is utilized in supervised learning techniques to enable the AI models to discover patterns and connections in the data. Following instruction, each model's performance is gauged using various measures including accuracy, precision, recall, F1-score, and ROC curve. This contrast helps to locate the top-performing model used next to integrate in the general system therefore enabling the most effective algorithm to be applied in classifying land transaction data.

Smart contract programming.

The suggested mechanism depends on smart contract creation. The Solidity programming language is used for developing smart contracts. meant to manage several land registration activities including those for registering buyers, sellers, and title transfer. Before being deployed, a vulnerability check using the Slither tool helps to confirm the validity and security of these intelligent contracts. This evaluation is essential to expose and reduce possible weaknesses and therefore improve the general resiliency of the property registration system.

Integration of blockchain systems

Deploying verified smart contracts on the Ethereum blockchain, especially the test network at Sepolia, completes the integration. This particular distribution ensures openness and integrity of land registration techniques. Furthermore, non-fraudulent transactional data is stored on the InterPlanetary File System (IPFS) and corresponding hash values are kept on the blockchain. By guaranteeing data integrity and immutability, this approach provides a reliable system to handle land ownership files.

Analyzing of the system; testing

To assess its performance, the system goes through a rigorous evaluation and testing process. Performance metrics vary, including artificial intelligence statistical ones such as precision, accuracy, recall, F1 score, and ROC curve. Furthermore measured in terms of transaction throughput and latency as well as gas cost usage for smart contract execution, blockchain scalability is also tested. Furthermore measured is the IPFS bandwidth consumed to store data. A cost estimate considers the savings that come from not storing on the blockchain and saving storage as well as from avoiding storage.

TABLE 1: FEATURE SIGNIFICANCE BASED ON THE KRUSKAL-WALLIS METHOD

Feature Name	Importance Score
Time Difference Between First and Last Transactions	374.0391
Average Time Between Received Transactions	72.3027
Total Transactions Required for Contract Creation	62.3562
Received Transactions	33.3540
Sent Transactions	32.3142
Average Value Sent	32.0028
Unique Addresses Sent	26.4828
Unique Addresses Received	6.8428
Average Minimum Between Sent Transactions	5.5799
Total ERC20 Transactions	5.0293
ERC20 Minimum Value Sent	4.1234
ERC20 Average Value Sent	4.0944
ERC20 Maximum Value Sent	3.0488
ERC20 Total Either Sent	3.6785
Max Value Sent	3.6573
Minimum Value Received	3.4480
Training Accuracy	83.19%

TABLE 2: FEATURE SIGNIFICANCE BASED ON CHI-SQUARED METHOD

Feature Name	Importance Score
Average Time Between Sent Transactions	741.9552
Average Time Between Received Transactions	740.5483
Total Transactions Required for Contract Creation	636.7954
Total Ether Sent	615.0080
Sent Transactions	614.8650
Index	614.86
Minimum Value Received	553.4257
Total Ether Balance	517.4098
Maximum Value Sent	510.5881
Unique Addresses Sent	478.6920
Average Value Sent	375.4685
Received Transactions	316.0090
Unique Addresses Received	272.5272
Minimum Value Sent	113.3185
Training Accuracy	92.86%

Table 3: feature significance based on anova method

Feature Name	Importance Score
Total Ether Sent	557.0533
Sent Transactions	514.3420
Total ERC20 Transactions	214.5823
Maximum Value Sent	417.9641
ERC20 Maximum Value Received	376.9784
ERC20 Total Value Received	345.9347
Total Transactions Required for Contract Creation	345.0733
Time Difference Between First and Last Transactions	469.5645
Unique Addresses Sent	310.6674
Total Ether Balance	298.7652
Average Value Sent	284.5691
ERC20 Unique Contract Address	232.7279
ERC20 Unique Token Name	232.1797
Average Minimum Between Sent Transactions	228.8379
ERC20 Unique Received Address	228.6403
Training Accuracy	97.74%

RESULT AND DISCUSSION

This part presents the implementation aspects, smart contract parameters, development tools and libraries, and the functions embedded in the proposed system's smart contracts. It also measures the performance of the system based on statistical measures such as training accuracy, precision, recall, F1-score, and log loss. Other discussions involve scalability, sustainability, and a comparison of block sizes.

I. Experimental setup and tools

The system to be proposed combines several smart contracts that have been built with The smart contracts are developed using the Remix Integrated Development Environment (IDE), which are coded in Solidity v0.8.16, are deployed to the Ethereum Blockchain (EB) network. The primary functions of the smart contracts are registerBuyer, registerSeller, requestLand, addLand, and

others. The contracts are built with the Solidity compiler v0.8.17 before being deployed onto the EB network with a gas allowance of 4,000,000 using the MetaMask wallet. Prior to deployment, security of such smart contracts is verified through the Slither analyzer tool, which detects vulnerabilities in source code and thereby protects against threats. InterPlanetary File System (IPFS) It is utilized for off-chain land registry storage to improve security and performance. further optimization. There are five machine learning models in the system that are used to process land transaction data: Light Gradient Boosting Machine (LightGBM), Extreme Gradient Boosting (XGBoost), Decision Tree (DT), Logistic Regression (LR), and Support Vector Machine (SVM). The outputs from these models, including blockchain scalability, ROC curve, log loss score, accuracy comparison, precision, recall, and F1-score, are plotted with the help of Python's Matplotlib library.

II. AI models ROC curve

The Receiver Operating Characteristic (ROC) curve visually represents the performances a classification model. The work compares LightGBM, XGBoost, Decision Tree, Logistic Regression, and Support Vector Machine on how well they can distinguish between classes. The ROC curve compares the True Positive Rate (TPR) to the False Positive Rate (FPR) across different classification thresholds.

- True Positive Rate (TPR) quantifies how well the model classifies actual positive instances.

- False Positive Rate (FPR) is the ratio of wrongly classified negative instances.

LightGBM performs better than the other models, with the highest TPR for a specific FPR, which reflects a better trade-off between sensitivity and specificity. This implies that LightGBM can identify positive instances with high accuracy while minimizing false positives. In comparison, XGBoost, Decision Tree, and Logistic Regression have comparable performance, whereas SVM has the worst classification accuracy. These findings identify LightGBM as the best-performing model out of the ones evaluated, hence the ideal one for fraudulent detection in the suggested land registration system.

III. Comparison of AI model accuracy

Accuracy for five classification models is gauged by this study. Accuracy captures the ratio of correctly labeled instances in a data set. LightGBM and XGBoost produced the best accuracy, doing better than the other models. This indicates that these two models were better in their ability to precisely classify data than DT, LR, and SVM, the least accurate being LR. LightGBM and XGBoost have better accuracy because they are able to support high-dimensional datasets and class imbalance problems well. These features render them suitable for tasks where high classification accuracy is paramount. According to these

results, LightGBM and XGBoost are the most appropriate models for classification problems in comparable applications where accuracy is considered a critical performance measure.

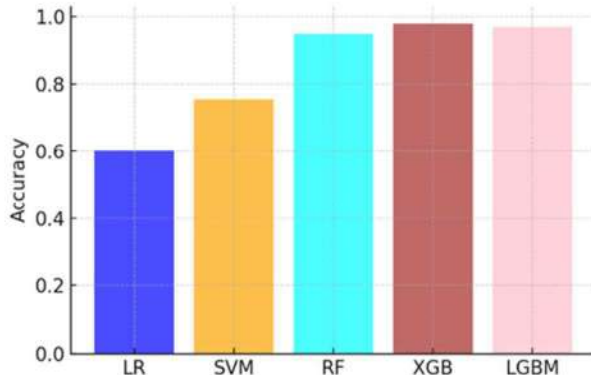


Figure 2. Accuracy comparison

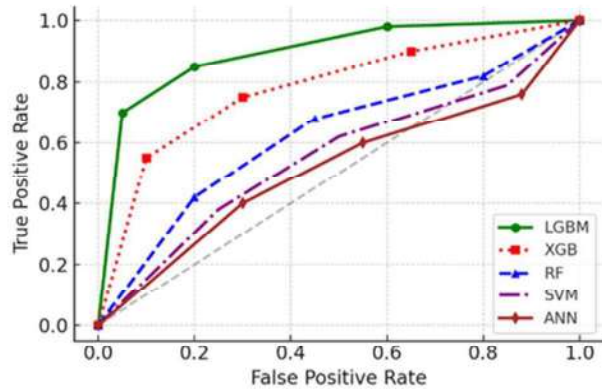


FIGURE 3. ROC CURVE

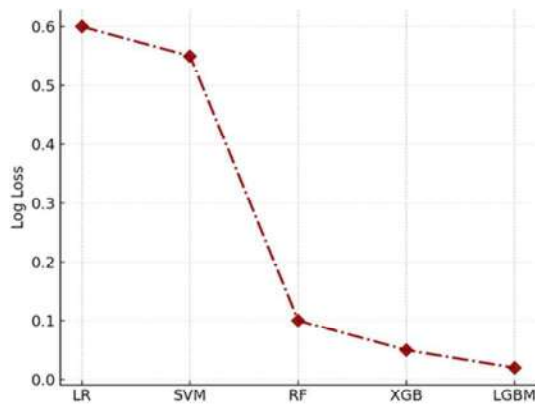


FIGURE 4. LOG LOSS

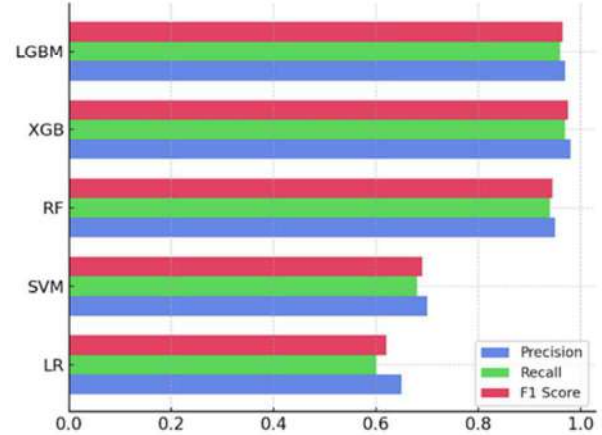


FIGURE 5. PRECISION, RECALL, AND F1 SCORE

IV. Comparison of AI model precision, recall, and F1-score

While accuracy is a key measure, it does not always accurately convey the performance of a classification model. To further support accuracy, this research also assesses precision, recall, and F1-score for all models. According to FIGURE 2d, LightGBM performed the best among all the other models in precision, recall, and F1-score. Precision conveys the fraction of accurately marked positive instances, while recall reveals the percentage by which actual positive cases are actually detected. The F1-score, the harmonic mean of precision and recall, offers a compromise between these two measures. DT and XGBoost produced similar results, and SVM and LR performed the least in these. LightGBM's high precision, recall, and F1-score can be explained by its capacity to handle imbalanced data and efficiently classify both positive and negative instances. These findings indicate that LightGBM is a very good option for classification problems in which precision and recall are vital.

V. AI model log loss score evaluation

Apart from accuracy and other performance indicators, log loss score was utilized to measure the AI models. Log loss is a commonly used metric for binary classification problems is assessing the quality of a model's probability estimates for various classes. Through the comparison of the log loss scores of the AI models, we learn how well they perform in giving precise probability estimates, dealing with imbalanced data, and staying efficient overall. LR had the highest log loss score, which means that it gave the least accurate probability estimates to the right class. Conversely, LightGBM recorded the lowest log loss value, indicating its best capacity for producing accurate probability estimates for every class. LightGBM's robust performance on log loss is due to its ability to handle imbalanced data efficiently while making very accurate probability predictions. These results highlight the need to test AI models using a variety of performance measures

instead of accuracy alone. taking into account metrics such as log loss, precision, recall, and F1-score, a more holistic evaluation of a model's strengths and weaknesses can be obtained. This method allows for improved decision-making when choosing the most appropriate model for a particular classification task.

VI. Blockchain scalability comparison

Blockchain scalability is crucial for processing transactions efficiently without compromising security and performance. It evaluates how well the blockchain can handle an increasing number of transactions.scales with rising transaction volumes and user activity without slowing down, getting congested, or being susceptible to security attacks. FIGURE 3 shows a comparison of the scalability of the proposed Ethereum-based system and a typical blockchain network. The y-axis is the number of transactions, and the x-axis is transaction time in milliseconds. As indicated in FIGURE 3, the system proposed has higher scalability than typical blockchain networks. The benefit arises from incorporating IPFS, which provides decentralized storage, lower costs, and higher data availability. IPFS achieves this by spreading files across several network nodes, thereby preventing threats such as censorship and single points of failure typical in centralized storage systems. Moreover, Ethereum's blockchain also utilizes technological innovations to enhance scalability, as opposed to conventional Bitcoin-based blockchains that have inherent scalability limitations in accommodating large-scale transactions. One of the salient features of the suggested system is that only genuine land records that are validated by the AI model are kept on the blockchain, and fraudulent data is rejected. This selective retention greatly minimizes computational overhead so that the blockchain only processes authenticated data. Consequently, response times are enhanced, and scalability overall is increased, as illustrated in FIGURE 3.

VII. IPFS bandwidth

In order to maximize storage efficiency and reduce costs, the system proposed uses IPFS rather than storing all data on Ethereum nodes directly. This method dramatically decreases bandwidth usage while enhancing data retrieval and storage speed. IPFS is locally installed and linked to a decentralized peer-to-peer network, where data is accessed through distinctive hash keys. shows the input and output bandwidth usage over a representative period of one minute. The findings reveal that the proposed system runs with significantly low bandwidth usage, which makes it an effective substitute for traditional blockchain storage techniques.

VIII. Smart contract implementation

The implementation of smart contract functionality in the Remix IDE for the secure land registry system operates

on the Ethereum-based public blockchain. These functions serve as a validation mechanism, ensuring that only valid data is recorded. in the blockchain. This is done to improve data accuracy, consistency, and reliability, which are critical to make their decision.Smart contracts on the blockchain consume computational resources, which have an associated gas cost. The proposed system operates under a Proof-of-Work (PoW) consensus mechanism, where miners execute smart contract transactions. Users must pay gas fees to compensate miners for their computational efforts. Gas costs are analyzed using the gas profiler tool in Remix IDE.The transaction and execution costs of key smart contract functions in gas units. Table 10 provides a breakdown of all implemented smart contract functions.Graph presents a comparison of gas consumption with and without AI integration. It can be seen that using AI significantly lowers both transaction and execution costs across all smart contract functions, increasing efficiency in the and registry system based on blockchain technology.

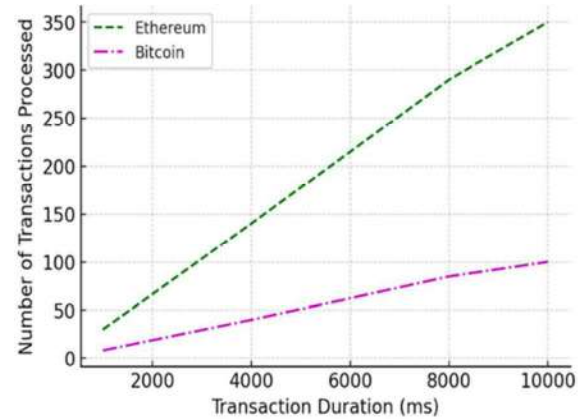


FIGURE 6. BLOCKCHAIN SCALABILITY COMPARISON

CONCLUSION AND FUTURE IMPROVEMENT.

Regarding land ownership and fraud identification in agriculture, the recommended land registration process addresses urgent issues. The initiative seeks to enhance the safety, transparency, and efficiency of land transactions through the use of blockchain technology.According the study, traditional land registration processes have weaknesses in fraud, insufficient openness, and transaction processing efficiency. The system fosters trust among stakeholders and promotes sustainable agriculture by providing a solid answer using state-of-the-art technologies. Furthermore, smart contracts reduce the risk of human errors by automating several activities and accelerate transactions. This clever approach usually not only preserves land rights but also secures the economic viability of the farming sector. The system might be expanded to allow real-time market analysis, automatic land valuation, and integration with other land

management systems. This would let people who are considering land deals and agricultural investments to use knowledge make proper judgments on land value trends. Including machine learning algorithms would also better predictive analytics since it would enable users to predict market changes and therefore adjust their plans as called for. Mobile app development could also improve the platform by allowing users to see property data and transaction capabilities on their phones. Furthermore, partnerships with banks could help buyers and sellers more easily find funding options, therefore promoting greater land market activity. Finally, the system might explore integration with devices to collect real-time data on land conditions, including soil health and crop yields, therefore even more improving the decision-making abilities of stakeholders in the farming industry.

REFERENCES

- [1]. Z. Iqbal, M. A. Khan, M. Sharif, J. H. Shah, M. H. U. Rehman and K. Javed, "An automated detection and classification of citrus plant diseases using image processing techniques: A review", *Comput. Electron. Agricult.*, vol. 153, pp. 12-32, Oct. 2018.
- [2]. I. Haider, M. A. Khan, M. Nazir, A. Hamza, O. Alqahtani, M. T.-H. Alouane, et al., "Crops leaf disease recognition from digital and RS imaging using fusion of multi self-attention RNet deep architectures and modified dragonfly optimization", *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 7260-7277, 2024.
- [3]. F. D. Rodrigues, J. M. Alonso and P. M. R. Castro, "SICAP—Cadastral information and property management system", *Proc. 17th Iberian Conf. Inf. Syst. Technol. (CISTI)*, pp. 1-6, Jun. 2022.
- [4]. S. N. Pereira, N. Tasnim, M. S. Rahman, S. N. Tumpa, M. M. Karim, R. S. Rizon, et al., "Multi-channel approach towards digitizing the land management system of Bangladesh", *Proc. IEEE Int. WIE Conf. Electr. Comput. Eng. (WIECON-ECE)*, pp. 68-71, Dec. 2018.
- [5]. E. Choudhury, M. Ridwan, M. A. Awal and S. Hosain, "A web-based land management system for Bangladesh", *Proc. 14th Int. Conf. Comput. Inf. Technol. (ICCIT)*, pp. 321-326, Dec. 2011.
- [6]. A. L. Shrivastava and R. Kumar Dwivedi, "Blockchain-based secure land registry system using efficient smart contract", *Proc. Int. Conf. Intell. Data Commun. Technol. Internet Things (IDCIoT)*, pp. 165-170, Jan. 2023.
- [7]. G. Kusuma, C. Rupa, S. Reshma and G. Rochana, "Secure storage of land records and implementation of land registration using Ethereum blockchain", *Proc. 3rd Int. Conf. Artif. Intell. Smart Energy (ICAIS)*, pp. 404-409, Feb. 2023.
- [8]. S. Dubey, D. Vinod, A. Gupta and R. K. Dwivedi, "Secure land registration management via Ethereum blockchain", *Proc. Int. Conf. Intell. Data Commun. Technol. Internet Things (IDCIoT)*, pp. 185-191, Jan. 2023.
- [9]. R. Sidharthan and V. R. Balasaraswathi, "Secured land registration using Ethereum blockchain and IPFS", *Proc. 3rd Int. Conf. Appl. Artif. Intell. Comput. (ICAAIC)*, pp. 1449-1454, Jun. 2024.
- [10]. N. Ncube, B. Mutunhu and K. Sibanda, "Land registry using a distributed ledger", *Proc. IST-Africa Conf. (IST-Africa)*, pp. 1-7, May 2022.
- [11]. M. I. Khalid, J. Iqbal, A. Alturki, S. Hussain, A. Alabrah and S. S. Ullah, "Blockchain-based land registration system: A conceptual framework", *Appl. Bionics Biomechanics*, vol. 2022, pp. 1-21, Feb. 2022.
- [12]. R. C. Suganthe, N. Shanthi, R. S. Latha, K. Gowtham, S. Deepakkumar and R. Elango, "Blockchain enabled digitization of land registration", *Proc. Int. Conf. Comput. Commun. Informat. (ICCCI)*, pp. 1-5, Jan. 2021.
- [13]. I. Mishra, A. Sahoo and M. V. Anand, "Digitalization of land records using blockchain technology", *Proc. Int. Conf. Advance Comput. Innov. Technol. Eng. (ICACITE)*, pp. 769-772, Mar. 2021.
- [14]. M. Nandi, R. K. Bhattacharjee, A. Jha and F. A. Barbhuiya, "A secured land registration framework on blockchain", *Proc. 3rd ISEA Conf. Secur. Privacy (ISEA-ISAP)*, pp. 130-138, Feb. 2020.
- [15]. R. Khan, S. Ansari, S. Sachdeva and S. Jain, "Blockchain based land registry system using Ethereum blockchain", *Xi'an Jianshu Keji Daxue Xuebao/J. Xi'an Univ. Archit. Technol.*, vol. 12, pp. 3640-3648, Apr. 2020.
- [16]. S. A. Gollapalli, G. Krishnamoorthy, N. S. Jagtap and R. Shaikh, "Land registration system using block-chain", *Proc. Int. Conf. Smart Innov. Design Environ. Manage. Planning Comput. (ICSIDEMPC)*, pp. 242-247, Oct. 2020.
- [17]. D. Shinde, S. Padekar, S. Raut, A. Wasay and S. S. Sambhare, "Land registry using blockchain—A survey of existing systems and proposing a feasible solution", *Proc. 5th Int. Conf. Comput. Commun. Control Autom. (ICCUBEA)*, pp. 1-6, Sep. 2019.
- [18]. P. Singh, "Role of blockchain technology in digitization of land records in Indian scenario", *IOP Conf. Ser. Earth Environ. Sci.*, vol. 614, no. 1, Dec. 2020.
- [19]. V. Aliyev, Ethereum Fraud Detection Dataset Version 1, Mar. 2021, [online] Available: <https://www.kaggle.com/datasets/vagifa/ethereum-frauddetection-dataset>.